Visualization as analyses

Readings for today

- Cairo, A. (2012). The Functional Art: An introduction to information graphics and visualization. New Riders. Chapters 1 & 3.
- Yanai, I., & Lercher, M. (2020). A hypothesis is a liability. Genome Biology, 21, 1

Topics

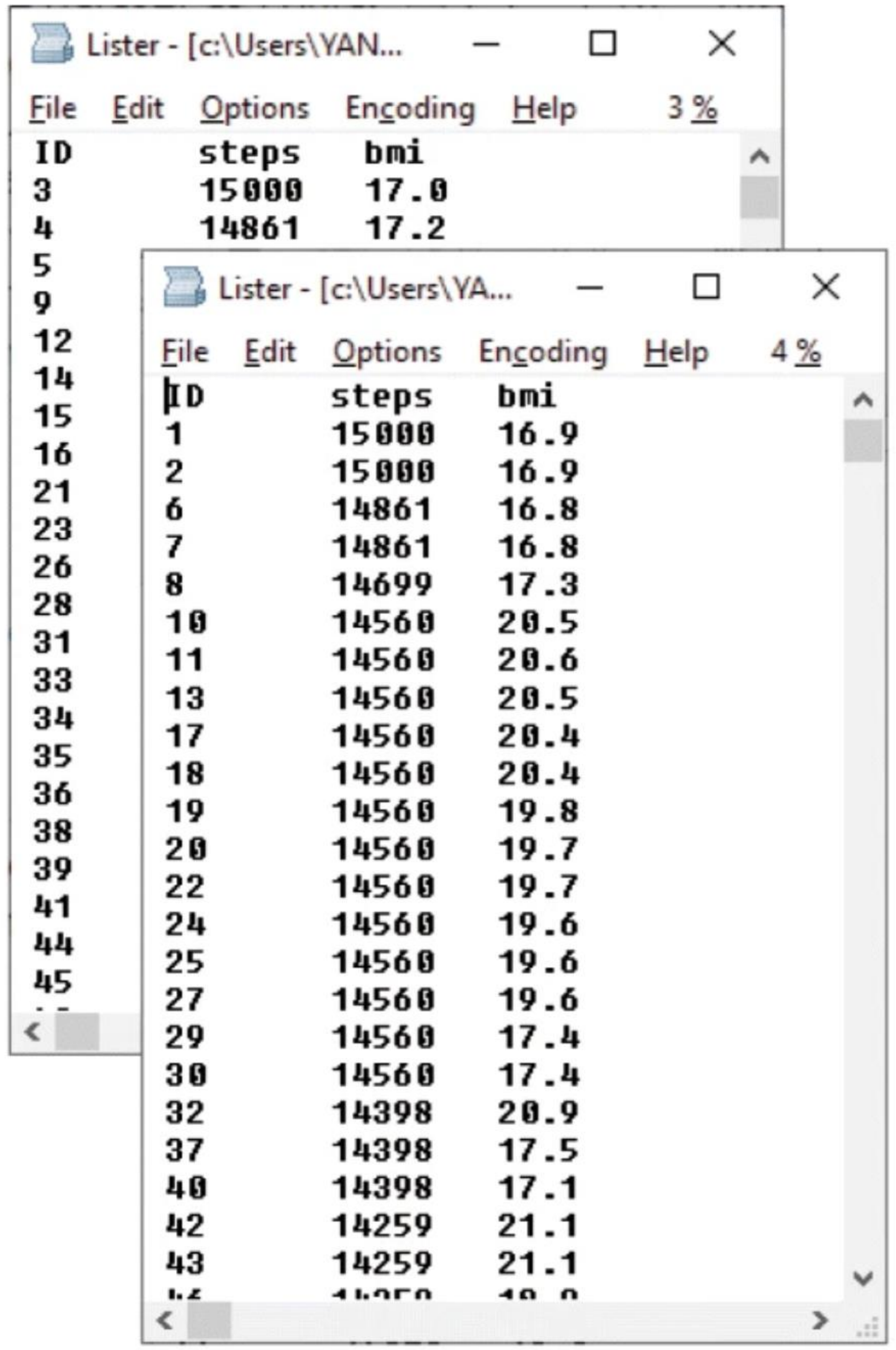
1. Goals of visualization

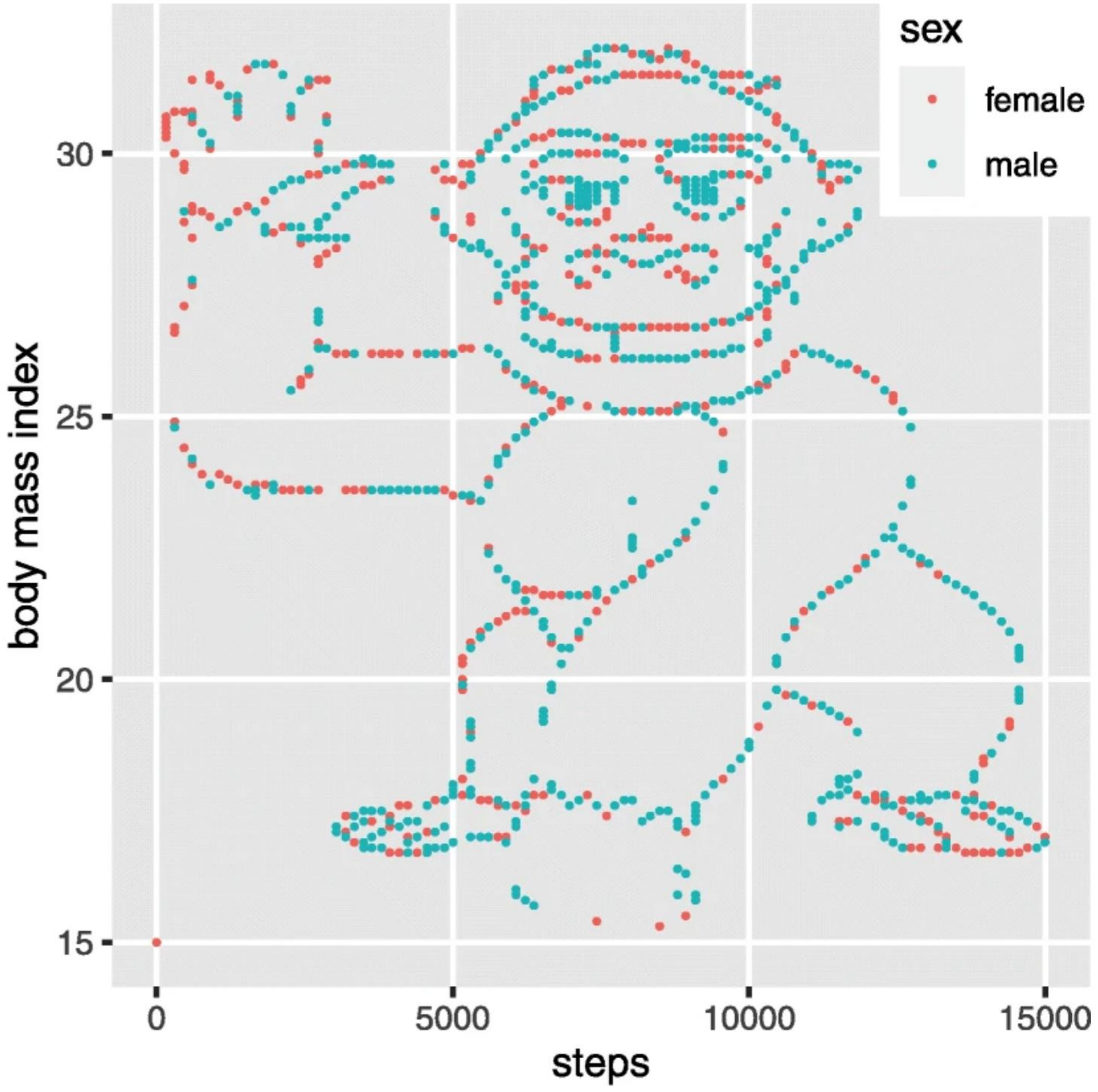
2. Balancing a paradox

Goals of visualization

Data to perception

Data table:





Visualization reveals structure in the data.

	Gorilla <u>not</u> discovered	Gorilla discovered
Hypothesis-focused	14	5
Hypothesis-free	5	9

(Yanai & Lercher 2020)

Graphical excellence

Goal: The efficient communication of complex quantitative ideas

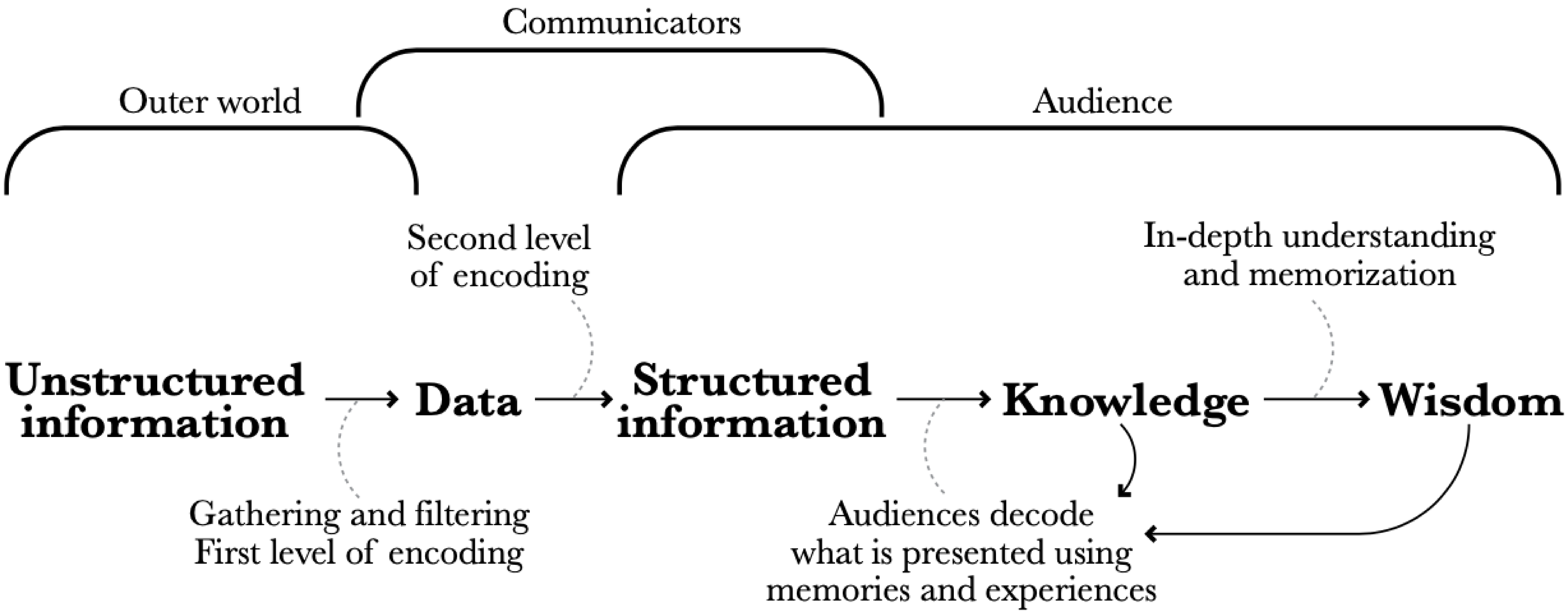
9 Properties of Graphical Excellence:

1. Show the data
2. Substance > method
3. Avoids distortion
4. Many numbers in a small space.
5. Coherent large data sets
6. Encourage visual comparisons
7. Many levels of detail
8. Clear purpose
9. Integrated with statistics.

"Graphics should reveal data in a meaningful & intuitive way." - Edward Tufte

(Tufte 1983)

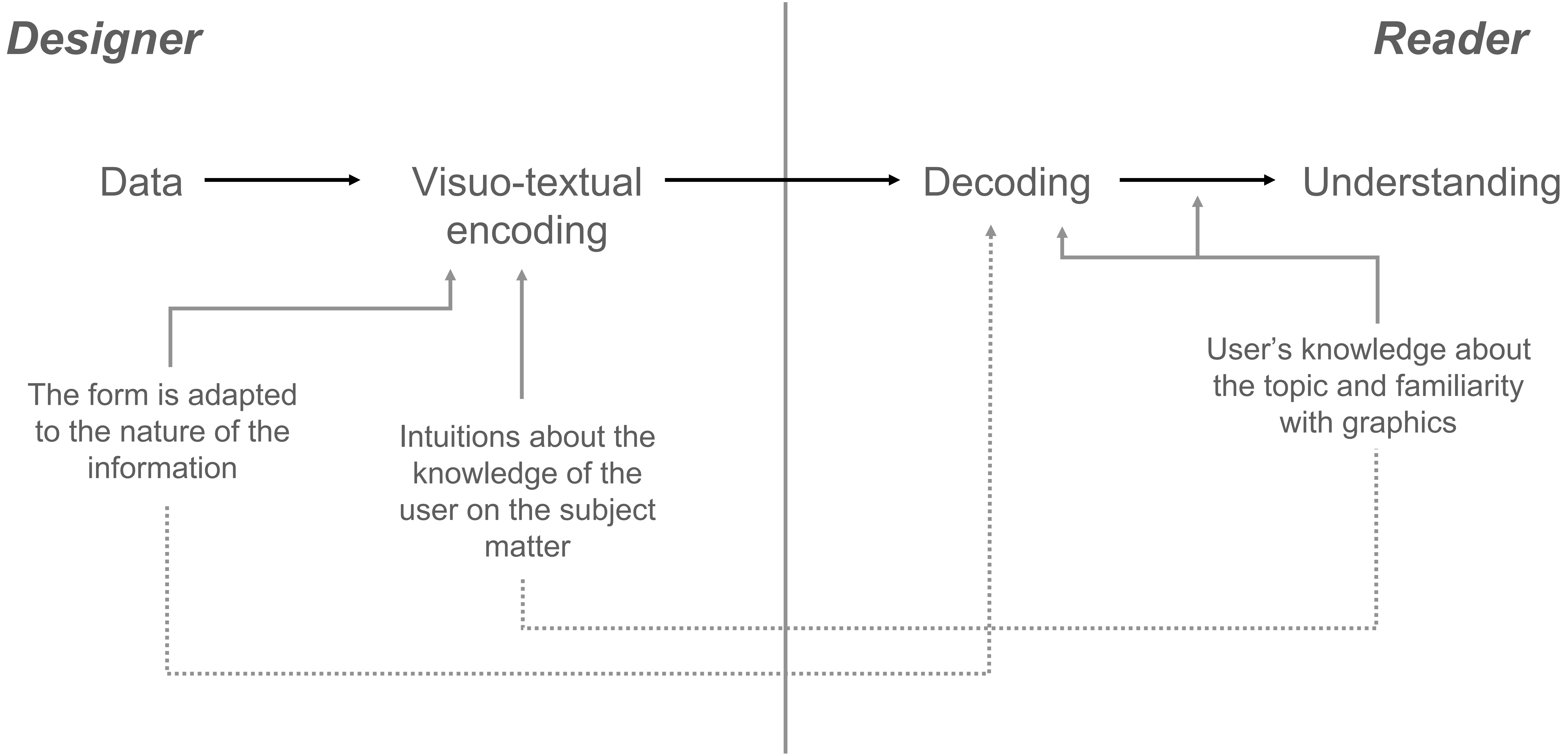
From visualization to wisdom



(Cairo 2012; Weinberg 2010)

Balancing a paradox

Think carefully of the decoding process



(Cairo 2012)

A struggle between competing goals

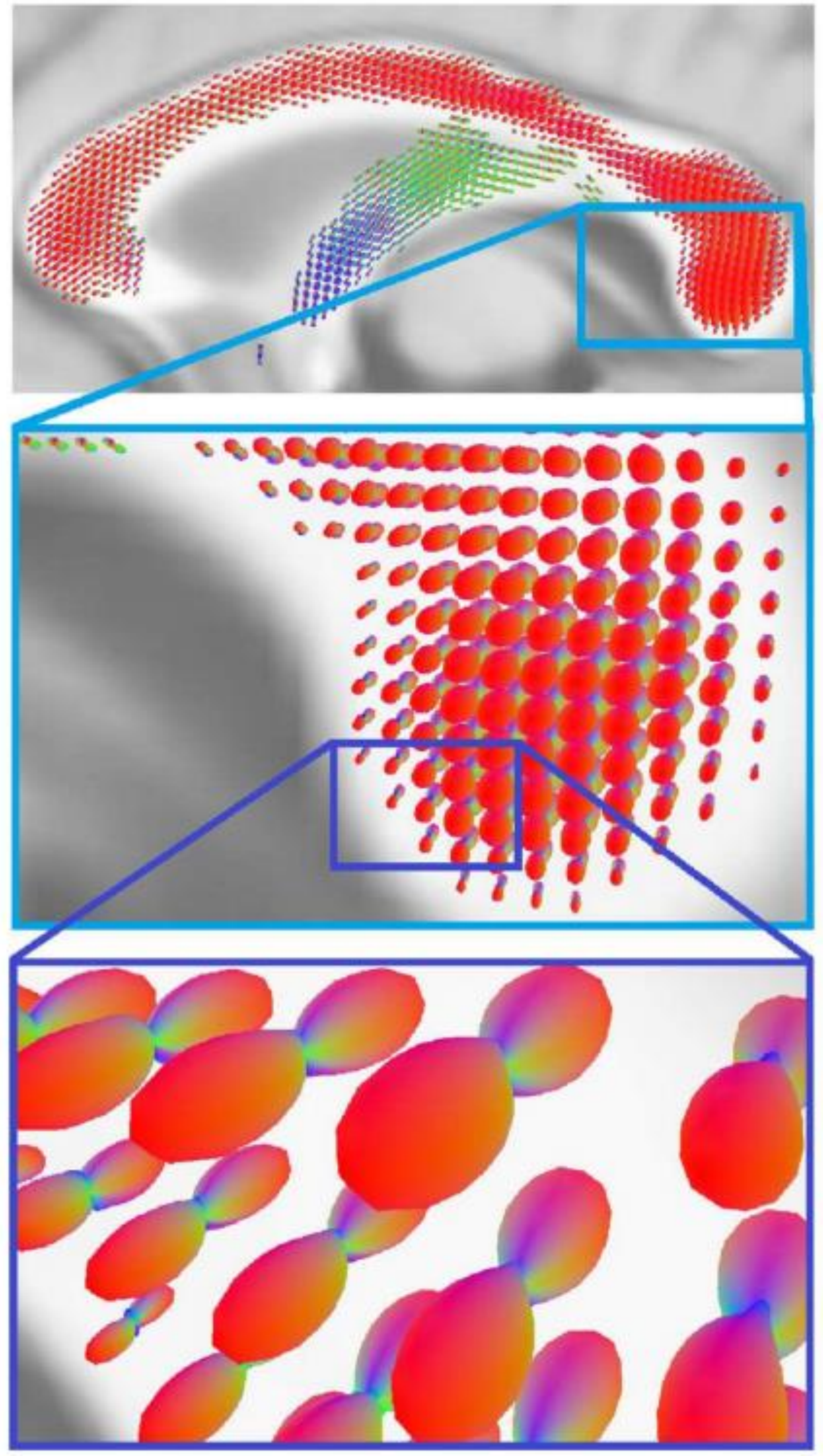
(Cairo 2012)

Abstraction vs. Figuration

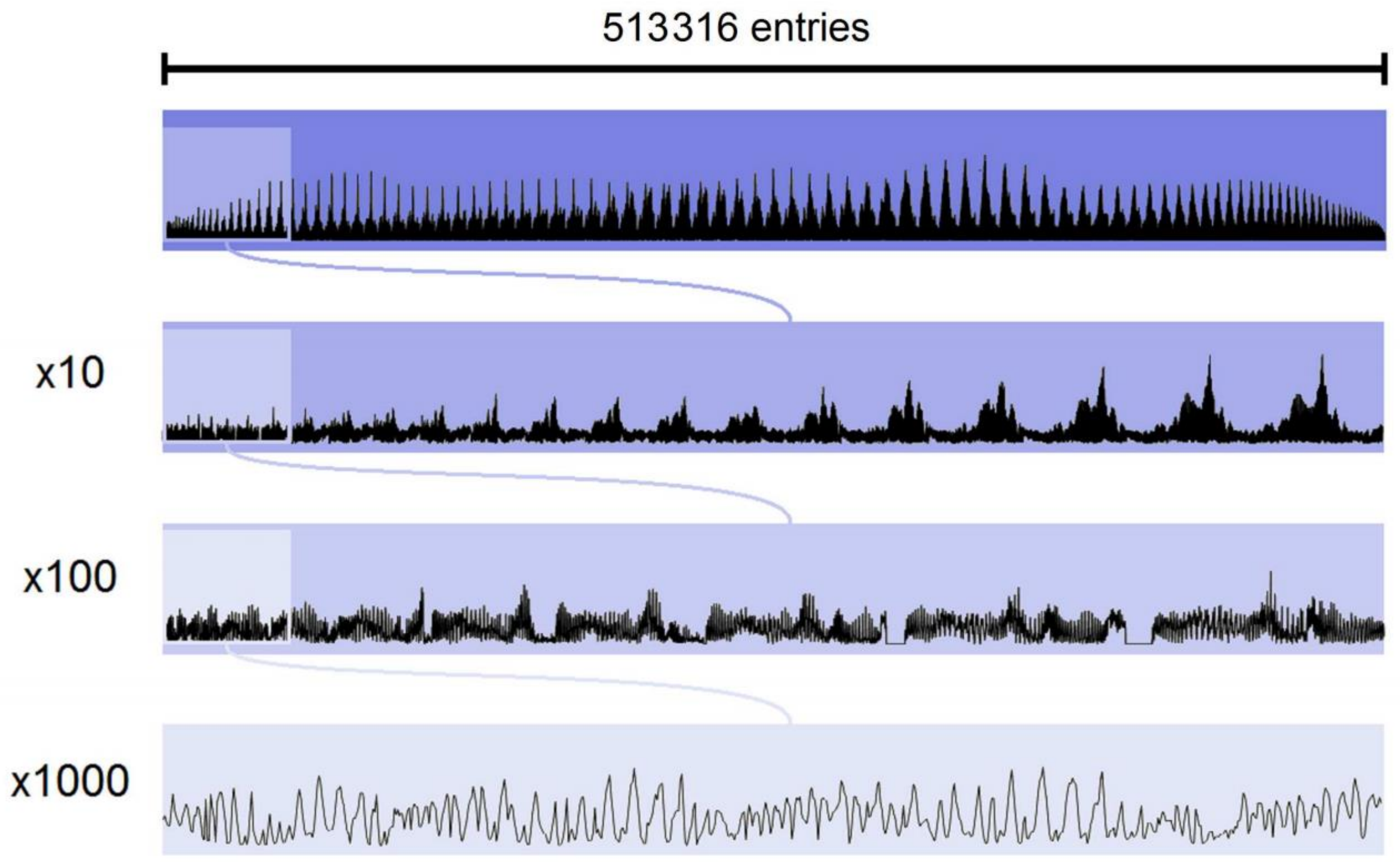
(Yeh et al. 2016)

The more distant the representation & its referent, the more abstract the visualization

Figurative



Abstract



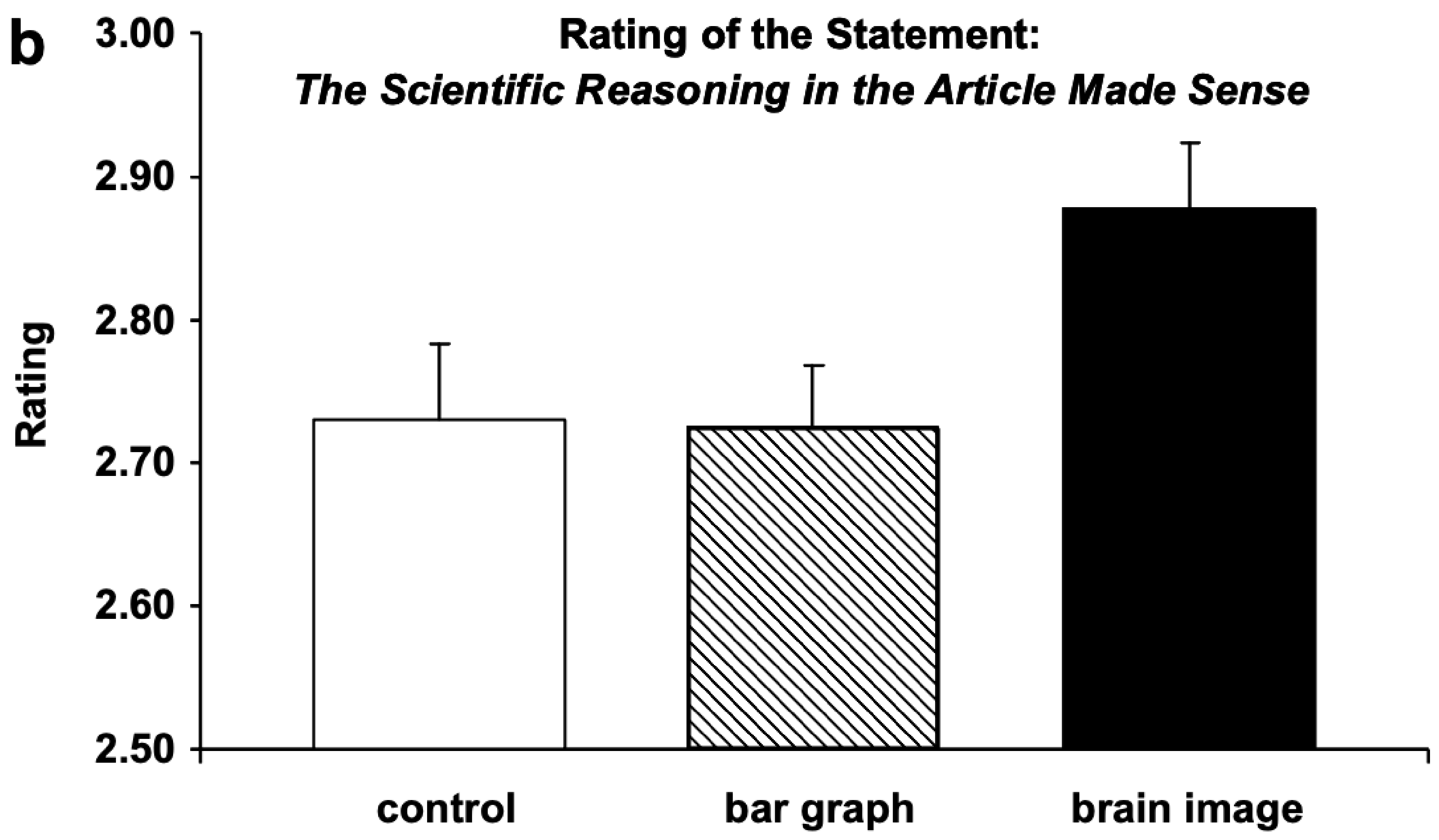
(Cairo 2012)

Functionality vs. Decoration

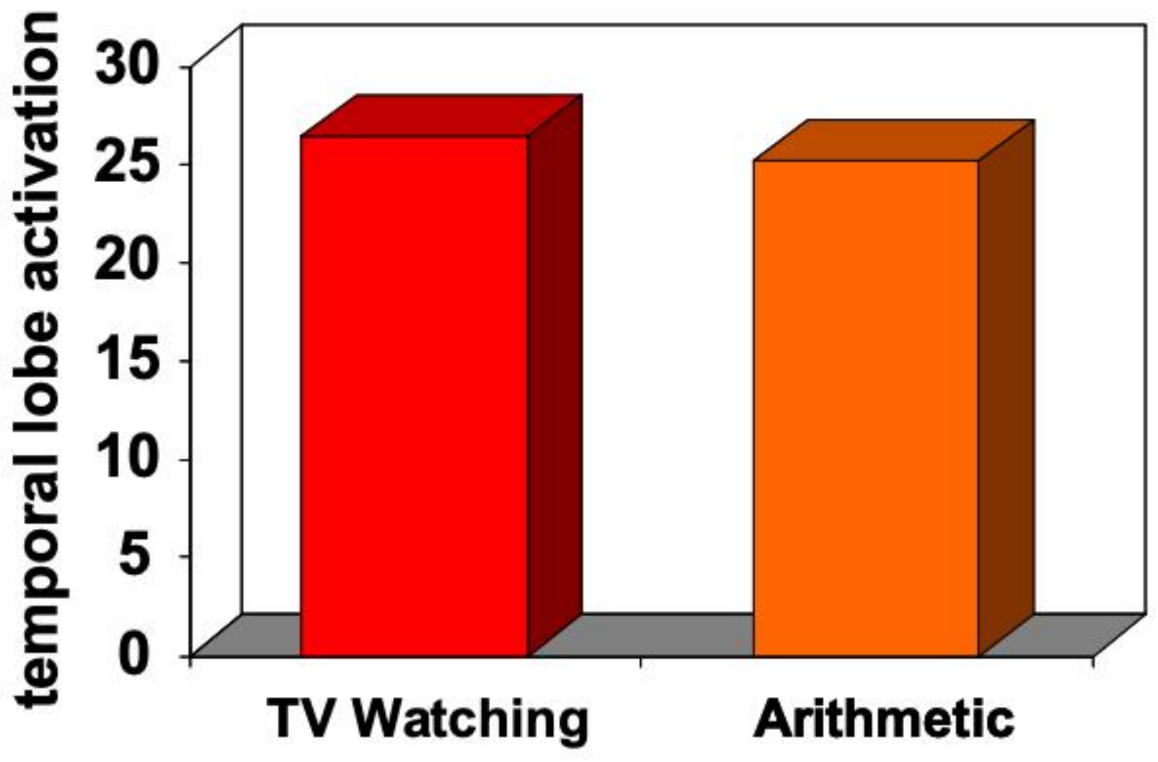
(McCabe & Castel 2008)

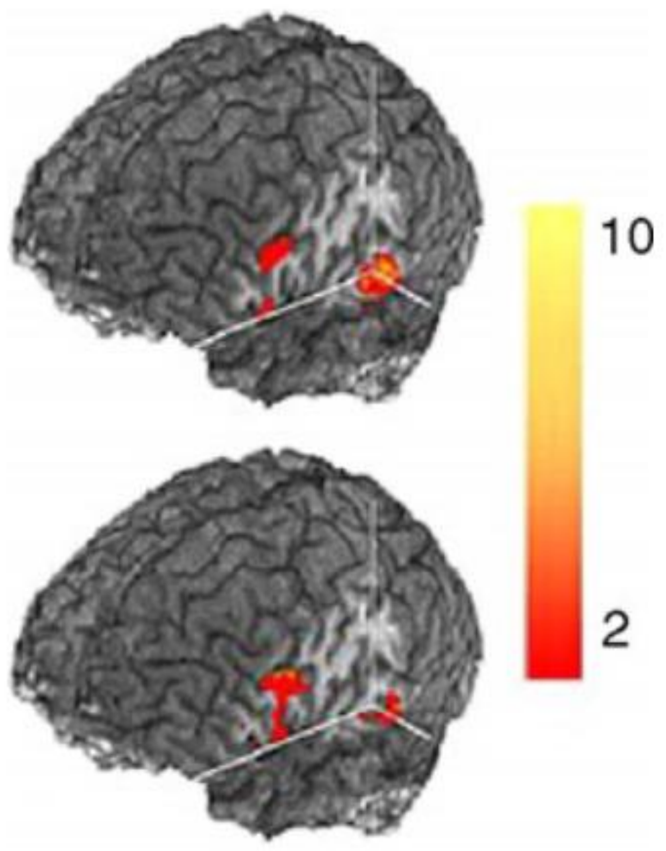
The inclusion of visual elements to grab attention, but are not directly relevant to enhancing comprehension, are decorative.

Functional



Decorative



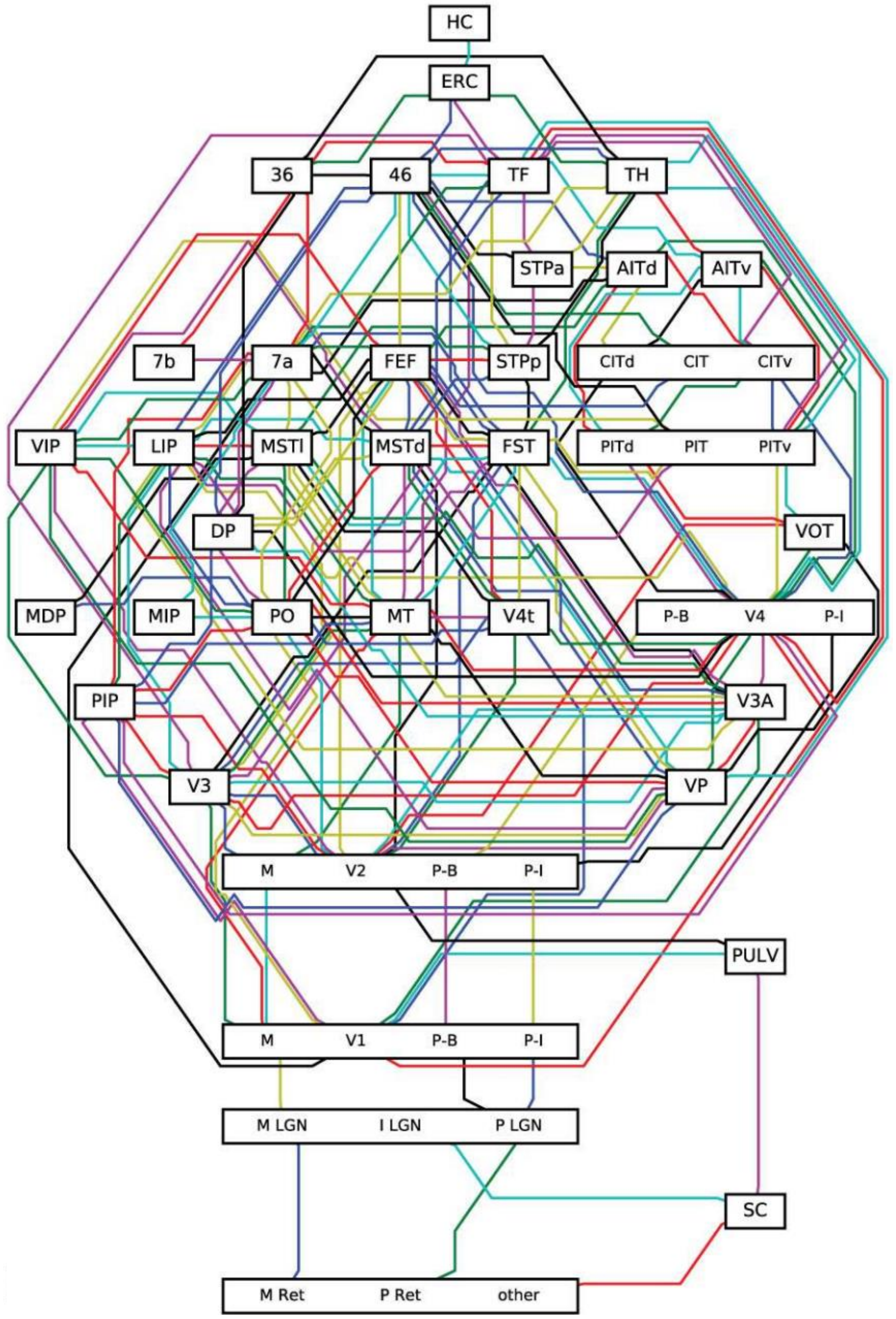


Density vs. Lightness

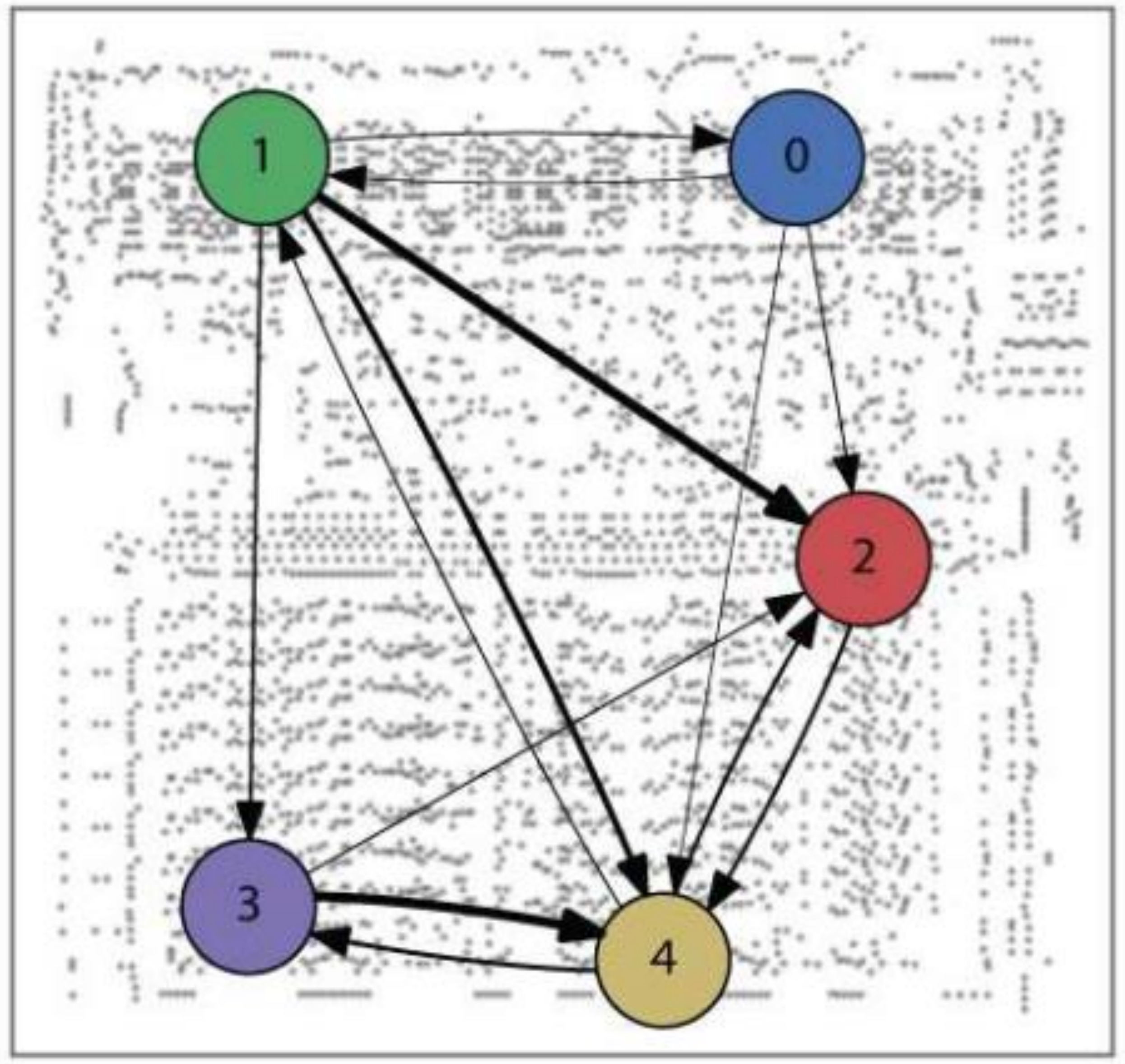
(Jonas & Körding 2017)

The greater the density of visual information, the more complex the demands to decode the information.

Dense



Light



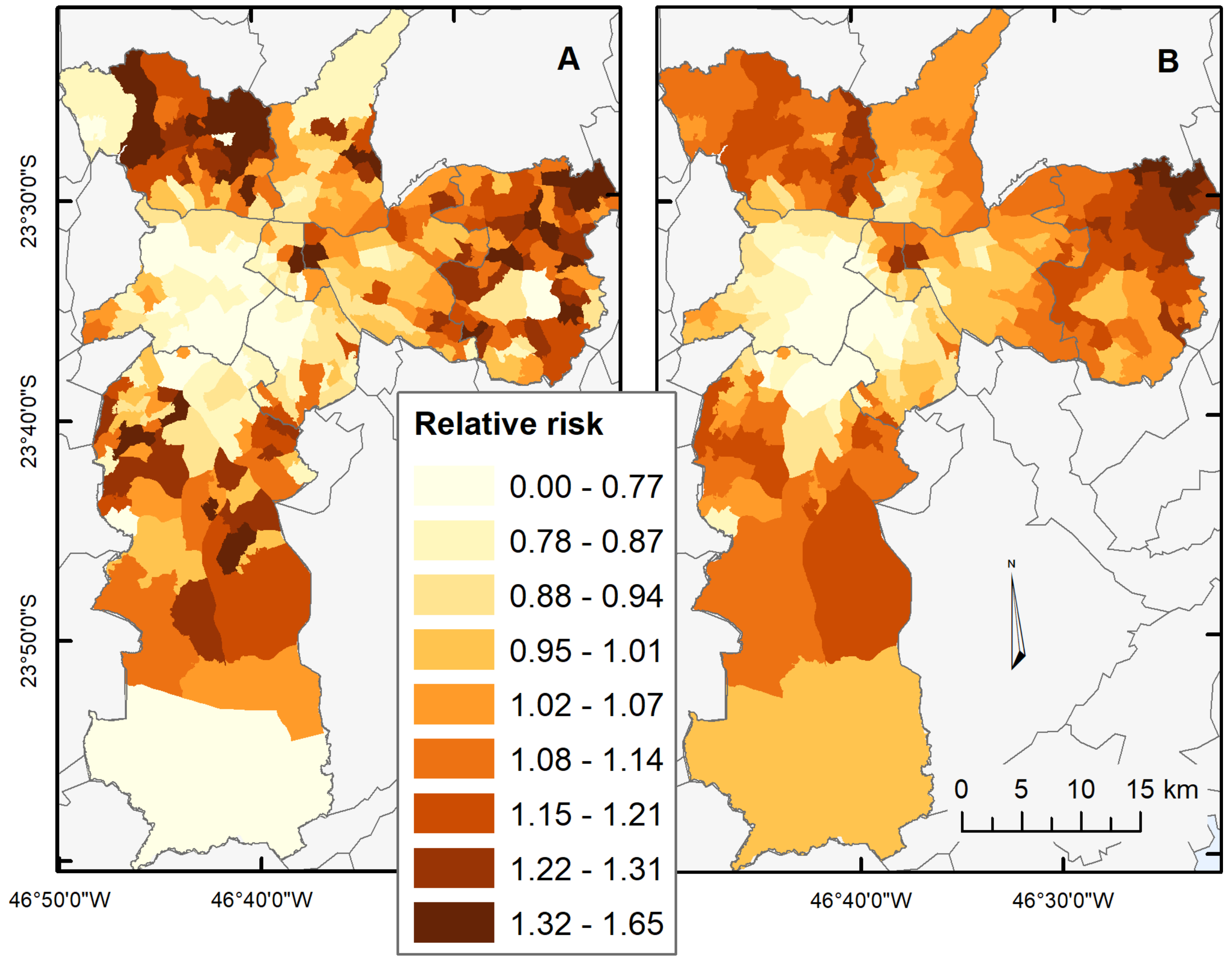
(Cairo 2012)

Multidimensional vs. Unidimensional

The more layers of depth, the greater the range of ways to decode information, but with increased decoding effort.

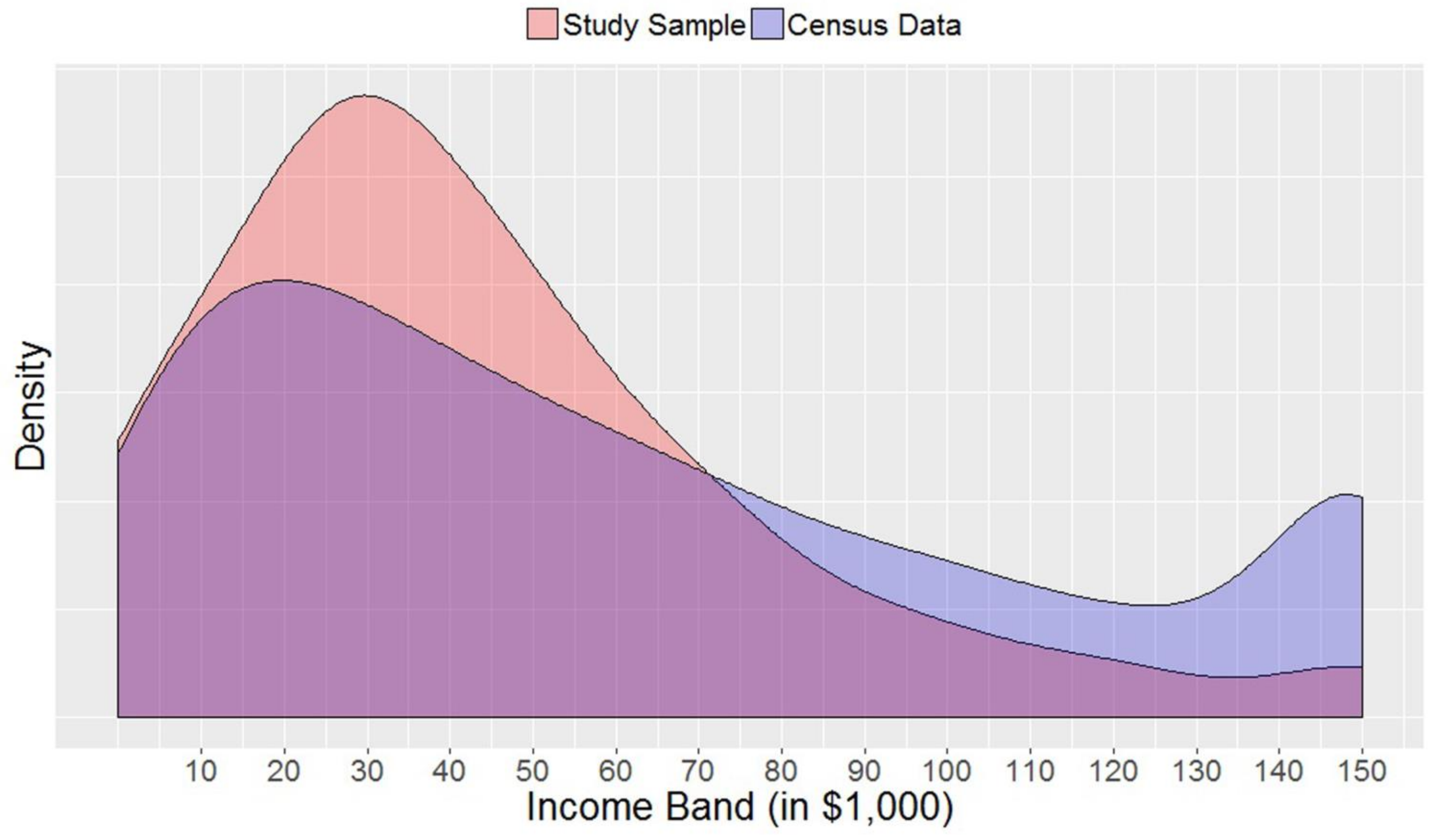
Multidimensional

(Vizeu Barrozo et al. 2020)



Unidimensional

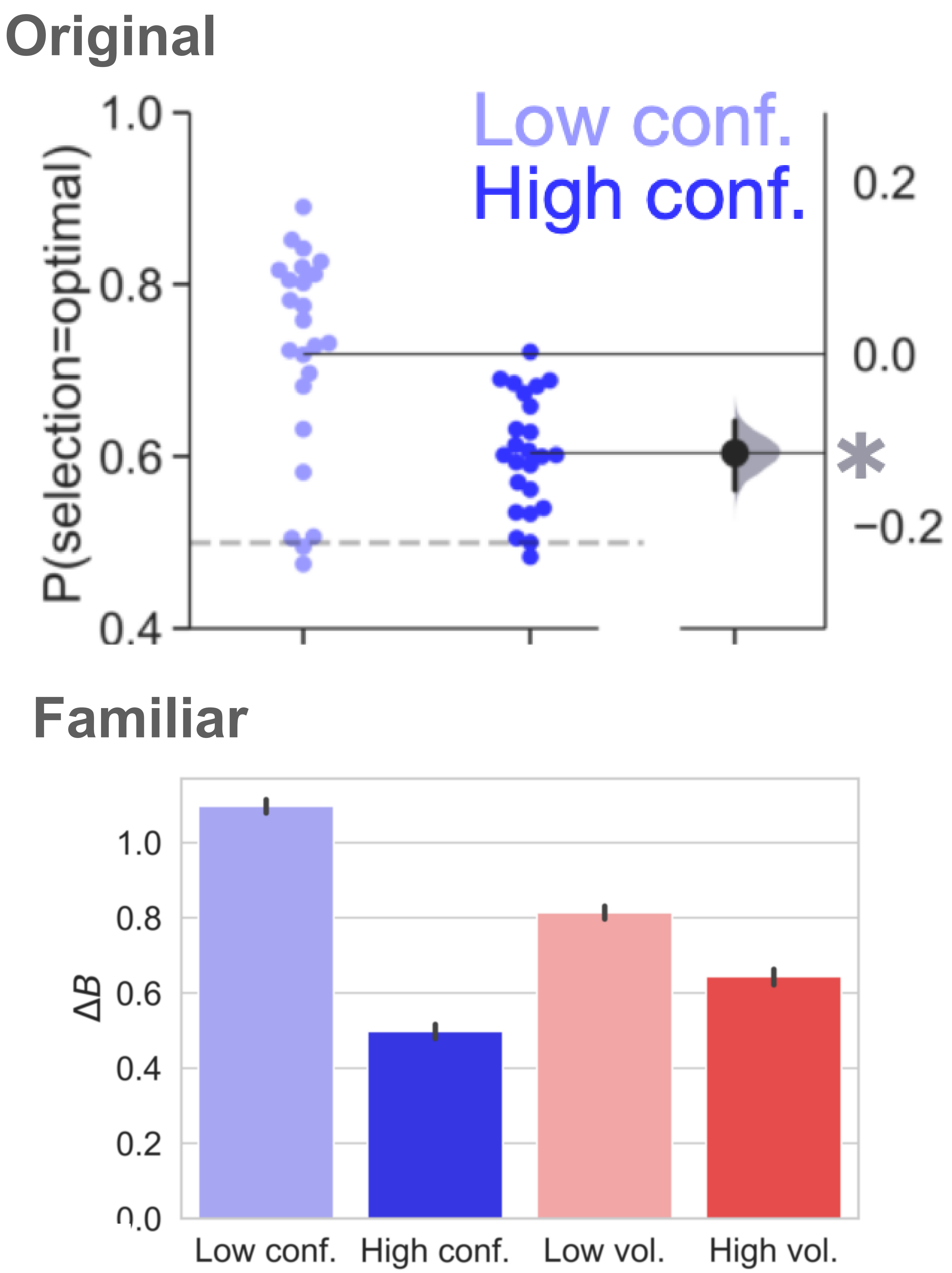
(Matz 2019)



Originality vs. Familiarity

More familiar plotting styles decrease the effort to encode information, but constrain ways of presenting relationships.

(Bond et al. 2021)



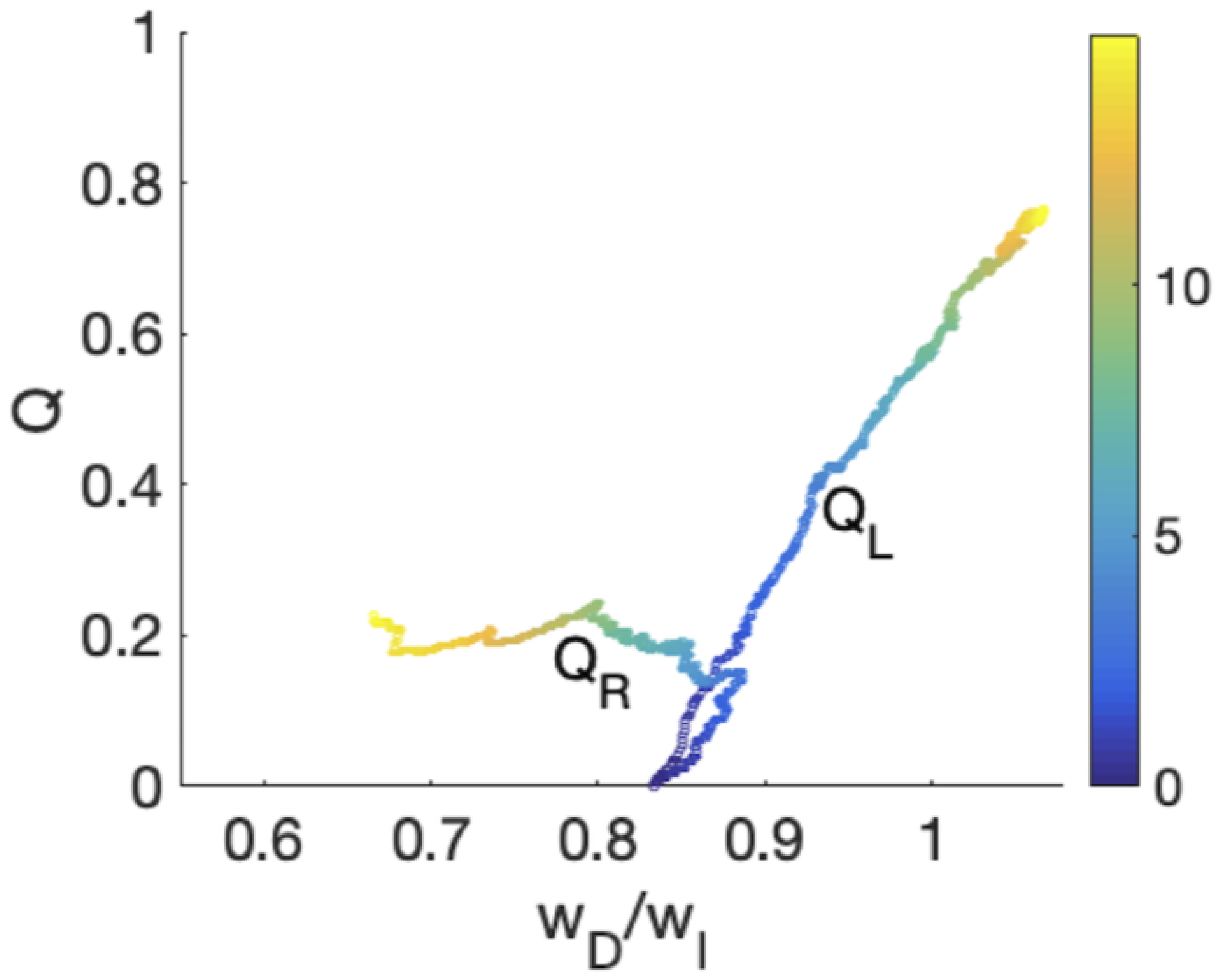
(Cairo 2012)

Novelty vs. Redundancy

Explaining many different things once (novelty) provides breadth, while explaining the same thing in different ways (redundancy) provides depth.

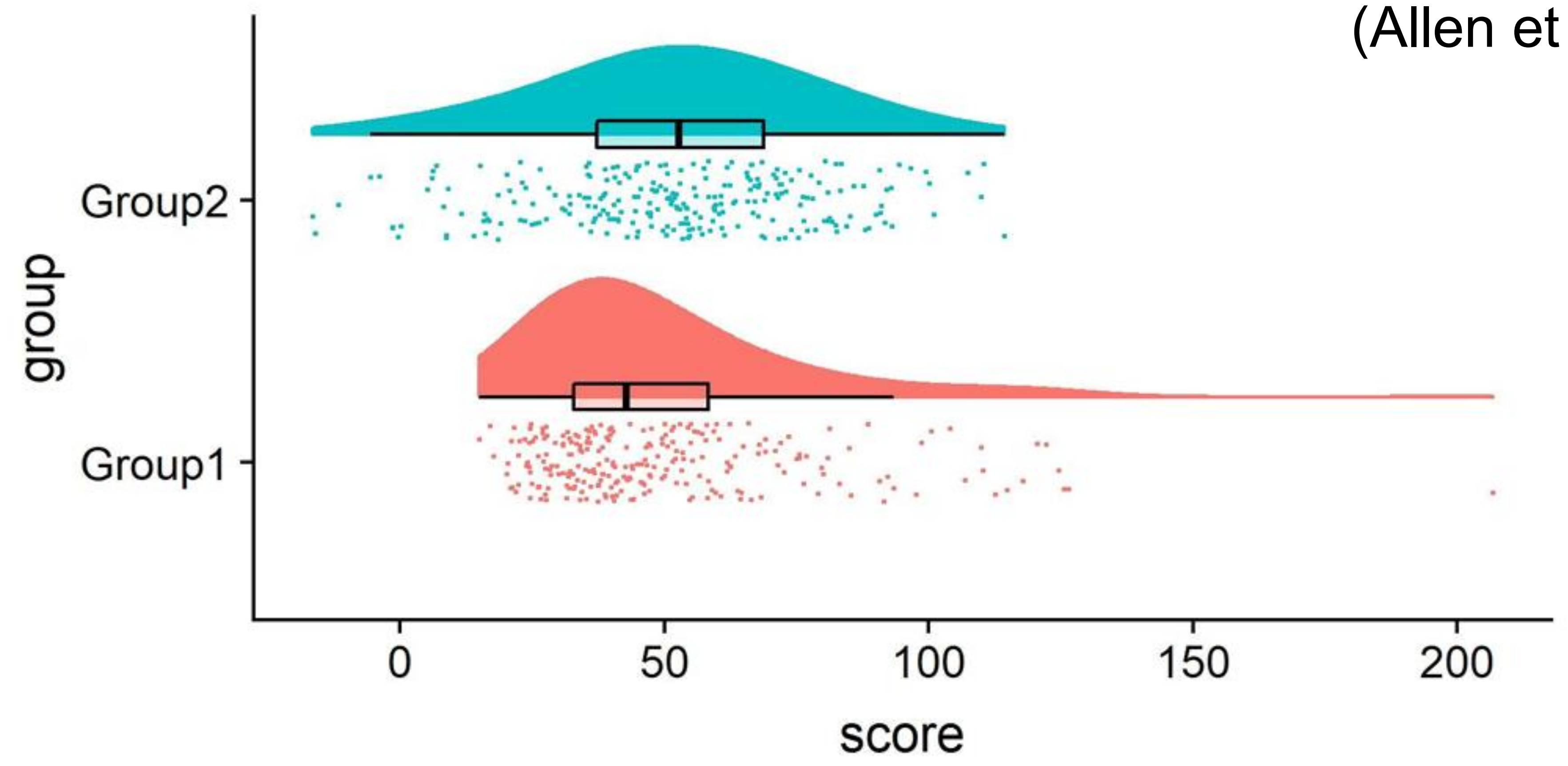
Novel

(Dunovan et al. 2019)



Redundant

(Allen et al. 2019)



(Cairo 2012)

A struggle between competing goals

(Cairo 2012)

Evaluating comprehension

What the reader should *immediately* understand

1. What is the figure’s central topic?
2. What phenomena & variables does the graphic show?
3. What changes does the graphic highlight in the data it represents?
4. Does the graphic present information in an objective manner, or does the author editorialize the content?

(Cairo 2012)

Take home message

- Visualization is your first and most effective tool for understanding and communicating relations in your data.
- Effective visual communication of relations in your data requires balancing multiple competing constraints.